

Hamzah Zuhrah

CYBERSECURITY STUDENT | IT SUPPORT | SOC & SECURITY OPERATIONS

Blacksburg, VA | (540) 835-1368 | szhamzah20@gmail.com | Portfolio | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Cybersecurity student at Old Dominion University (GPA 3.95, expected May 2027) and Virginia Tech industrial engineering graduate with hands-on IT support experience. Experienced in Windows and macOS troubleshooting, account access, networking, ticket documentation, Python security engineering, and clear support for nontechnical users. Seeking SOC, security operations, and IT security internships.

TECHNICAL SKILLS

Security: Incident response, log analysis, network security, threat modeling, access control, system hardening, phishing analysis
Systems & Networking: Linux/Kali Linux, Windows, macOS, TCP/IP, DNS, DHCP, Nmap, virtual machines, Wi-Fi troubleshooting
Programming & Tools: Python, Bash, Git, GitHub, automated testing, TLS, AES-256-GCM, scrypt, structured audit logging
IT Support: Tier 1 support, ticketing systems, account access, password resets, software installation, device configuration, user education

PROFESSIONAL EXPERIENCE

Information Technology Specialist | Gateway Homes, Inc. Jan 2025 - Jan 2026

- Supported computer, Wi-Fi, and audiovisual setup and maintenance while troubleshooting account access and general user issues.
- Provided calm, practical technical support during day-to-day operations and events for users with varied technical backgrounds.

Information Technology Help Desk Technician | Virginia Tech Jan 2025 - Jun 2025

Blacksburg, VA | Part-time

- Diagnosed hardware, software, account, printer, Wi-Fi, and application issues for students, faculty, and staff across Windows and macOS.
- Completed password resets, software installation, and device configuration; documented resolutions, escalated complex cases, and coached users on phishing and password safety.

Industrial Engineering Intern | Moog Space and Defense Jan 2022 - Aug 2022

Blacksburg, VA | Internship

- Analyzed aerospace production workflows and operational data, identified inefficiencies, and communicated process findings with cross-functional teams.

SELECTED CYBERSECURITY PROJECTS

Secure File Transfer System v2 | Python Security Engineering 2026

- Re-engineered a socket prototype with client-side AES-256-GCM, verified TLS 1.2+, salted scrypt authentication, rate limiting, audit logs, and per-user storage boundaries.
- Designed a versioned length-prefixed protocol and validated tamper, traversal, authentication, transfer-integrity, and real TLS workflows with 11 automated tests.

BreachCoach | Incident Response Tabletop Simulator 2026

- Built three branching scenarios with 27 scored decisions and printable after-action reviews informed by NIST SP 800-61r3 and CISA guidance.

EDUCATION

B.S. Cybersecurity | Old Dominion University Expected May 2027

GPA: 3.95

Bachelor of Engineering, Industrial Engineering | Virginia Tech May 2025